not derive the correct data types correctly based on the document's field values. Incorrect data types lead to setbacks during search operations.

Fortunately, Elasticsearch allows us to create indexes to meet our requirements by letting us configure and instantiate them explicitly as we require. Before we jump into developing custom indexes, we need to know about index configurations, discussed in the next section.

INDEX CONFIGURATIONS

Every index is made of a configuration consisting of mappings, settings, and aliases, regardless of whether it was created automatically or explicitly. We covered mappings in chapter 4; we recap here and present two other configurations:

- *Mappings*—Mapping is the process of creating a schema definition. Data that is stored usually has multiple data types associated with its fields, such as `text`, `keyword`, `long`, `date`, and so on. Elasticsearch consults the mapping definitions to apply appropriate rules for analyzing incoming data before storing it for efficient and effective searching. For example, the following snippet sets the mapping for the `cars_index_with_sample_mapping` index:

```
PUT cars_index_with_sample_mapping
{
  "mappings": {
    "properties": {
      "make":{
        "type": "text"
      }
    }
  }
}
```

We can issue the `GET cars_index_with_sample_mapping/_mapping` command to fetch the schema for our newly created `cars_index_with_sample_mapping` index.

- *Settings*—Every index comes with a set of configuration settings, such as number of shards and replicas, refresh rate, compression codec, and others. A few settings (called *dynamic settings*) can be tweaked on a live index at runtime. Other settings (*static settings*) are applied to an index in a non-operational mode. We look at these two types shortly. The next listing configures an index with a few settings.

Listing 6.3 Creating an index with custom settings

```
PUT cars_index_with_sample_settings
{
  "settings": {
    "number_of_replicas": 3,
    "codec": "best_compression"
  }
}
```

Creates the index

Applies the settings